



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**RESIT EXAMINATION
JUNE 2023**

Course : EC3247 (Cloud Architecture)

Time: 2.00 pm – 5.00 pm
(3 hours)

Lecturer: Muhammad Hafez Mohd Ariffin

Date: 28 September 2023

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **2** printed pages.
(excluding the front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. List **FOUR (4)** common characteristics of cloud computing. **(4 MARKS)**
 - b. Explain deployment procedures of cloud computing. **(6 MARKS)**
 - c. Explain any **THREE (3)** roles of open source software in cloud computing. **(6 MARKS)**
 - d. List any **FOUR (4)** levels of technical architecture. **(4 MARKS)**

2.
 - a. List any **FOUR (4)** examples of Platform as a Service (PaaS). **(4 MARKS)**
 - b. Explain the following dimensions of scalability in cloud computing:
 - i. Application scalability
 - ii. Technology scalability
 - c. What is system High Availability (HA) in the software model? **(4 MARKS)**
 - d. Explain the causes of information leakage, integrity violation, denial of service and improper authentication to cyberspace. **(6 MARKS)**

3.
 - a. Explain horizontal and vertical Software as a Service (SaaS). **(6 MARKS)**
 - b. Explain any **TWO (2)** adoption challenges of SaaS. **(6 MARKS)**
 - c. Discuss the following virtualized infrastructure manager (VIM) features: **(8 MARKS)**
 - i. Interface to public cloud
 - ii. Virtual networking
 - iii. Dynamic resource allocation
 - iv. Virtual clusters

4. a. Explain any **THREE (3)** features of Infrastructure as a Service (IaaS) model. (6 MARKS)
- b. Explain automatic scaling in SaaS model. (4 MARKS)
- c. Write about the Platform as a Service (PaaS) solution. (10 MARKS)
5. a. Describe **THREE (3)** types of cloud integration. (6 MARKS)
- b. Explain the following cloud deployment models in enterprise cloud computing: (6 MARKS)
- i. Community clouds
 - ii. Managed clouds
 - iii. Hybrid clouds
- c. Discuss any **FOUR (4)** data center security solutions in cloud computing. (8 MARKS)

-END OF QUESTION PAPER-